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Statement

Graduate from University of Cambridge with a passion for high-level coding, looking to help build outstanding tools and experiences. I'm a great team player, and I'm also keen to take the lead where appropriate. I'm a good communicator, both in-person and in writing; I'm precise and diplomatic in defining goals.

Education

University of Cambridge 2019 - 2022

Computer Science (BA) - 2.1

Lancaster Royal Grammar School (A-Levels 2017 - 2019)

A*/A*/A*/A* - Computer Science / Maths / Further Maths / Physics

Work

Triage App Programmer

- 2023-2024**
- Worked with a junior doctor to produce a prototype app for assisting with triage in ER & GP waiting rooms
 - Identified inefficiencies and redundancies from patient and doctor perspectives
 - Modelled problems with use of surveyed data and expert opinions to identify how an app could alleviate issues
 - Iteratively refined the UX with feedback from professionals
 - Created using Django

Inventory Developer & Server Maintainer

- Eggcup 2015 - 2021**
- Eggcup is a surplus food charity. As Eggcup scaled, it required a Food Inventory that could deal with irregular stock, and also required minimal training for use by volunteers.
 - Created an internal web-app using Django, Javascript and MySQL. It provided an easy, familiar interface for stock input/output; item normalisation to coalesce identical items; a flexible overview to help determine what food should be used when; customizable statistical summaries for management to provide to sponsors.
 - Refined the app over time, combining written feedback with in-person usability testing, further tailoring it to the charity's unique needs.
 - Provided maintenance and support over SSH and in-person, for the server device, inventory app and wordpress website.

Multimodal Interaction Group intern

- University of Glasgow 2018**
- Developed a proof-of-concept VR car demo, driven by "physically" interacting with the steering wheel via VR controllers rather than joysticks, to assist the group's research into motion sickness.
 - Wrote easy to understand public summaries of research papers for the group website.

Languages

- Python
- C++
- Javascript
- C#
- HTML
- SQL
- Java
- Svelte

Tools/Frameworks

- Unity
- Qt
- Flask
- Django
- Git
- pygame
- Mirror
- Netcode for Gameobjects

Projects

This is a highlight of the programming projects I've worked on. All were self-started, aside from Riff and Model Maker. They've all been valuable for developing experience, though with the knowledge gained I would certainly approach some of them differently now.

RogueFrame in particular challenged my preconceptions on multiplayer game networking, and also taught me the hard way why scriptable objects are great..

Group

- 2024** • **"Artifaction"**: "Artifaction" is a WIP social deduction game that aims to avoid player elimination (and its associated pitfalls), made in Unity with 1 other developer. I've worked on every part of the game, but had a focus on base systems and integrating external APIs (Vivox and Steam Audio).
- 2021** • **SpellCast**: 2D 3v3 proof of concept for an objective-based arena game, made in Unity with 2 others. I created the underlying systems used throughout the game, such as the base weapons scripts, and also had a strong hand in the weapon balance philosophy. Entry for NSE Games Innovation Challenge 2021
- **Riff**: 6-person coursework project developed during lockdown, aiming to provide a way for musicians to "jam" together online while bypassing the issue of latency. I worked on the front-end: designed the main page; implemented the UI; integrated the other members' components into the UI. A presentation with a demo and more details can be found at youtu.be/pg3UFd0-l8I

Solo

- 2023** • **Arkshot Mods**: Enables network-synced customisation of gameplay values for the Unity game Arkshot. Written using BepinEx, with a mix of C# hooks and CIL transpilers.
- 2022** • **RogueFrame**: WIP 3P shooter/roguelite in Unity, inspired by Risk of Rain 2 and Warframe.
- 2021** • **IncrementalBackup**: An incremental backup engine implemented in Java, aimed at use with minecraft. It allows backups to be much more frequent, by comparing the current state with the old one and only storing what has changed. The main engine is completely generic, and can have custom handlers for different file types implemented.
- 2017** • **Model Maker**: A-Level Computer Science project, implemented with Qt in C++. Produces demonstrations of a sub-set (focusing on friction) of A-level mechanics principles with custom saveable models.
- **3D Noughts And Crosses**: Implemented in pygame with a custom 3D renderer, with support for unlimited players and variable grid sizes.
- **MEGANAUGHTSANDCROSSES**: A game of noughts and crosses, where each grid is also a game of noughts and crosses. Implemented in pygame (with a text fallback option), and has LAN support.

Other Experience and Achievements

- **Netcraft AQA A Level Computing Award**: Top 50 results in AQA Computer Science.
- **Programming Society**: Co-founded, led and ran a school society dedicated to providing extra-curricular python teaching, as well as assisting student hobby projects.
- **Mentoring**: Provided mentoring to a Year 8 student, overcoming their struggles in Computer Science & Physics at LRGS, volunteered in an online programming teaching community.
- **Developments in VLSI Design Talk**: Researched, prepared & presented a 25 minute talk on very large scale integration in microprocessors for Churchill College, University of Cambridge. Received an award for "an outstanding talk".
- **Overwatch Commentator at Cambridge vs Oxford e-sports Varsity 2023-2024**: <https://youtu.be/JvjkFmFBgZ4> and <https://www.twitch.tv/videos/2127447965>. Quickly summarized live information and provided analysis in-person and on-stream.
- **Totally Normal High School**: Play-tested and balanced mechanics for Cambridge University Roleplaying and Tabletop gaming Society's in-house TTRPG. Refined my Quality Assurance and diplomatic client communication skills.
- **Vecter**: Worked with the developer by providing: numerous bug reports; and feedback which helped steer the direction of the game. I also later moderated the community discord server.